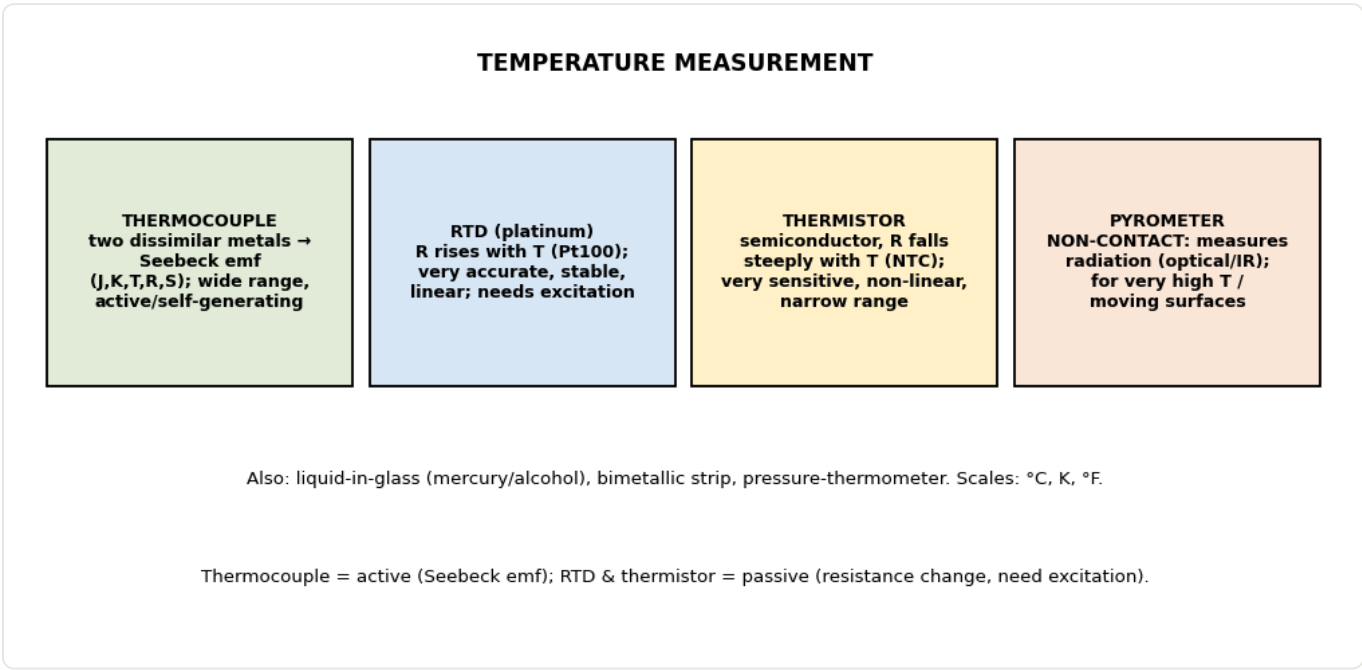
UNIT 9 · TOPIC 5(A)

Measurement of Temperature and Pressure

Syllabus topic: Measurement of temperature, pressure, strain, force, torque, power, sound level and vibrations

Temperature

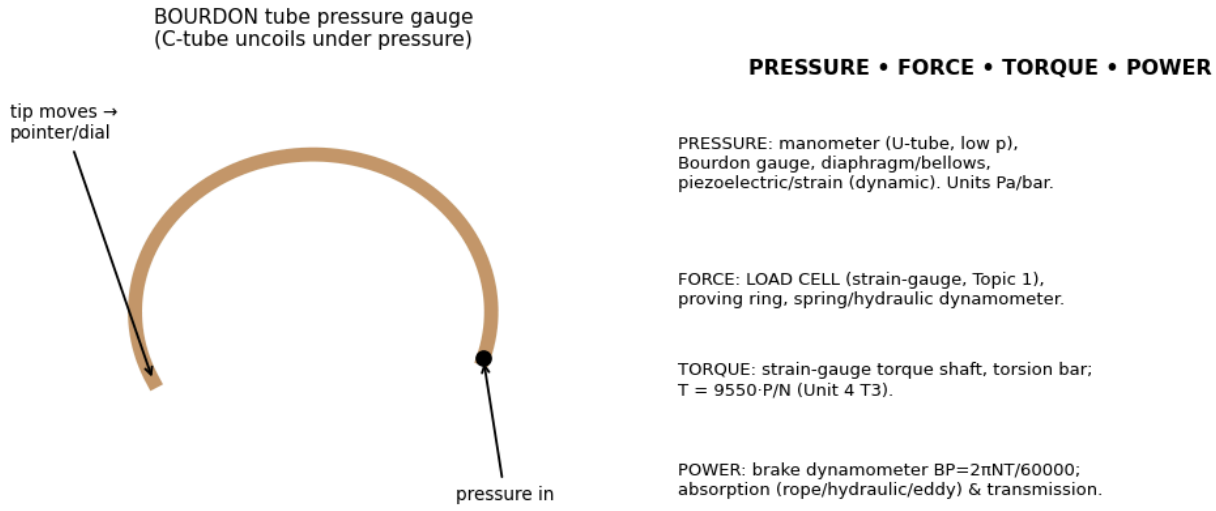


Temperature-measuring instruments

Instrument	Principle	Features
THERMOCOUPLE	Two dissimilar metal wires joined at a junction generate an emf ∝ junction temperature (SEEBECK effect)	Active (self-generating), rugged, wide range; needs a reference (cold) junction
RTD (Resistance Temperature Detector)	Usually PLATINUM — resistance rises nearly linearly with temperature (Pt100 = 100 Ω at 0 °C)	Very accurate and stable; passive (needs excitation)
THERMISTOR	Semiconductor — resistance FALLS steeply with temperature (NTC)	Very sensitive, but non-linear and narrow-range
PYROMETER	NON-CONTACT — infers temperature from emitted thermal RADIATION (optical/radiation/IR)	Very high temperatures; moving or inaccessible surfaces
Others	Liquid-in-glass (mercury/alcohol), bimetallic strip, pressure (filled-system) thermometers	—

Thermocouple type	Metals / use
J	Iron–constantan
K	Chromel–alumel — the most used, up to ~1200 °C
T	General low-temperature
R, S	Platinum-based — high temperature

Pressure



Bourdon-tube gauge and the pressure/force/torque/power instruments

Instrument	Use
MANOMETER (U-tube liquid column)	Low pressures and differentials
BOURDON-TUBE gauge	The common mechanical gauge — a curved C-tube uncoils as internal pressure rises, moving a pointer
Diaphragm and bellows elements	Moderate pressures
Piezoelectric or strain-gauge transducers	DYNAMIC / high-speed pressure

Units: Pa, kPa, bar. Distinguish absolute and gauge pressure.

Practice MCQs

Q36. A thermocouple works on the:

(a) Seebeck effect
(b) Photoelectric effect
(c) Hall effect
(d) Doppler effect

Q37. A thermocouple is a/an ___ transducer:

(a) Passive
(b) Active (self-generating)
(c) Digital
(d) Optical

Q38. The most commonly used general-purpose thermocouple is type:

(a) J (b) K (Chromel-Alumel)
(c) T (d) S

Q39. An RTD is usually made of:

(a) Copper only (b) Platinum
(c) Silicon (d) Carbon

Q40. In an RTD, as temperature rises the resistance:

(a) Falls (b) Rises (nearly linearly)
(c) Stays constant (d) Oscillates

Q41. A thermistor is made of a ___ and its resistance ___ with temperature (NTC):

(a) metal, rises (b) semiconductor, falls steeply
(c) ceramic, constant (d) gas, rises

Q42. Non-contact temperature measurement of a hot/moving surface uses a:

(a) Thermocouple (b) Pyrometer (radiation)
(c) Manometer (d) Rotameter

Q43. Pt100 means a platinum RTD with resistance:

(a) 100 Ω at 0 °C (b) 100 Ω at 100 °C
(c) 1000 Ω (d) 10 Ω

Q44. A U-tube manometer is best for measuring:

(a) Very high pressure (b) Low pressures / differentials
(c) Temperature (d) Flow rate only

Q45. The common mechanical pressure gauge using a curved tube that uncoils is the:

(a) Manometer (b) Bourdon gauge
(c) Thermistor (d) Rotameter

Q46. Dynamic (rapidly varying) pressure is best measured with a:

(a) Bourdon gauge (b) Piezoelectric/strain transducer
(c) Manometer (d) Dead-weight tester

Answers

Q36 — (a) Seebeck effect — *Thermocouple — Seebeck emf.*
Q37 — (b) Active (self-generating)
Q38 — (b) K (Chromel-Alumel) — *Type K most common.*
Q39 — (b) Platinum — *RTD — platinum.*
Q40 — (b) Rises (nearly linearly) — *RTD resistance rises nearly linearly.*
Q41 — (b) semiconductor, falls steeply — *Thermistor — semiconductor, R falls steeply.*
Q42 — (b) Pyrometer (radiation) — *Pyrometer — non-contact radiation.*
Q43 — (a) 100 Ω at 0 °C — *Pt100 = 100 Ω at 0 °C.*
Q44 — (b) Low pressures / differentials
Q45 — (b) Bourdon gauge — *Bourdon C-tube uncoils under pressure.*
Q46 — (b) Piezoelectric/strain transducer

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney; measuring-instruments course notes (as given in the source notes, July 2026); standard fuel-testing methods (ASTM/IS); NPTEL measurement lectures.